

# Treatment of Herpes Simplex Infections

## *Disseminated or Severe Herpes Infections*

The treatment of choice for disseminated or severe herpes infections is intravenous **acyclovir** at a dose of 5 to 10 mg/kg every 8 hours. For life-threatening HSV infections, including encephalitis, some experts recommend higher doses of acyclovir (15 to 20 mg/kg intravenously every 8 hours). In neonatal herpes, the recommended dose is 20 mg/kg of intravenous acyclovir every 8 hours.

## *First Episodes of Genital HSV-2 Infections*

Oral **acyclovir**, **famciclovir**, and **valacyclovir** are all effective in accelerating healing, resolving symptoms, and reducing viral shedding during initial genital herpes episodes. Compared with placebo: - Acyclovir reduces time to healing from 16 to 12 days - Duration of pain decreases from 7 to 5 days - Constitutional symptoms resolve faster, decreasing from 6 to 3 days

Valacyclovir has been shown to be equivalent to acyclovir in treating primary episodes. However, antiviral therapy during the initial episode does not reduce the frequency of future recurrences, likely because HSV establishes latency very early—within hours of inoculation and before symptoms appear.

## *Recurrent Genital Herpes Episodes*

Treatment of recurrent genital herpes with acyclovir, famciclovir, or valacyclovir shortens: - Healing time from 7 to 5 days - Viral shedding duration from 4 to 2 days - Symptom duration from 4 to 3 days (compared to placebo)

Valacyclovir and acyclovir have been shown to be equivalent in efficacy. Famciclovir is expected to perform similarly, though direct comparative data may be limited.

## *Suppressive Therapy for Frequent or Complicated Recurrences*

Long-term suppressive therapy with **acyclovir** or its analogues (e.g., valacyclovir, famciclovir) is the most effective strategy for individuals with frequent or severe recurrences. Since genital herpes is typically non-progressive in immunocompetent individuals and recurrence frequency may naturally decline over time, it is advisable to consider an annual "treatment holiday" to reassess the ongoing need for suppressive therapy.

## *Antiviral Use in Late Pregnancy*

Suppressive antiviral therapy during the third trimester (starting at 36 weeks' gestation) has been proposed to reduce the risk of neonatal herpes by preventing genital herpes recurrences at delivery. While studies suggest benefits in reducing clinical recurrences, cesarean deliveries due to herpes, and viral shedding at birth, formal large-scale trials are lacking due to the rarity of neonatal herpes. Therefore, while promising, further research is needed before definitive recommendations can be made.

## *Orolabial HSV Infections*

Antiviral treatment is less commonly indicated for orolabial herpes than for genital infections. However, **primary HSV gingivostomatitis in children should be treated with oral acyclovir**. The pediatric regimen is: - **Acyclovir suspension**: 15 mg/kg orally five times daily for 7 days

Initiation within 3 days of symptom onset reduces the duration of oral and extra-oral lesions, fever, and overall illness.

## **Box 193-1**

### **Recommended Regimens for the Treatment of Herpes Simplex Infections**

| DISEASE          | CLINICAL SITUATION | ADULTS                                   | PEDIATRIC | DURATION                              | COMMENTS                                        |
|------------------|--------------------|------------------------------------------|-----------|---------------------------------------|-------------------------------------------------|
| Orofacial herpes | Primary infection  | Acyclovir 200 mg orally five times daily | —         | 7–10 days or until symptom resolution | Intravenous acyclovir for severely ill patients |

| DISEASE | CLINICAL SITUATION | ADULTS                                                                                                                              | PEDIATRIC | DURATION | COMMENTS |
|---------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|
|         |                    | Acyclovir 400 mg orally three times daily<br>Valacyclovir 1000 mg orally twice daily<br>Famciclovir 250 mg orally three times daily |           |          |          |